

Revision (Lab 1 and Lab 2)

Scenario Based Question (DDL & DML)

A private engineering college wants to computerize its **Student Academic Management System**. The system should store information about **students**, **departments**, and **courses**. You are assigned as a database engineer to design and manipulate the database using MySQL.

Given Requirements

- Each student belongs to one department
- Each course is offered by one department
- Proper constraints must be applied to maintain data integrity

Part A: Database & Table Design (DDL)

Q1. Create a database named engineering_college.

Q2. Use the newly created database.

Q3. Create a table department with the following structure:

Column Name	Data Type	Constraints
dept_id	INT	PRIMARY KEY
dept_name	VARCHAR(50)	NOT NULL, UNIQUE
block	VARCHAR(20)	NOT NULL

Q4. Create a table student with the following structure:

Column Name	Data Type	Constraints
student_id	INT	PRIMARY KEY

name	VARCHAR(50)	NOT NULL
email	VARCHAR(50)	UNIQUE
gender	VARCHAR(10)	
dob	DATE	
dept_id	INT	FOREIGN KEY references department(dept_id)

Q5. Create a table course with the following structure:

Column Name	Data Type	Constraints
course_id	INT	PRIMARY KEY
course_name	VARCHAR(50)	NOT NULL
credit	INT	CHECK (credit > 0)
dept_id	INT	FOREIGN KEY references department(dept_id)

Q6. Display the structure of all the tables created.

Q7. Add a new column phone of type VARCHAR(15) to the student table.

Q8. Modify the gender column size to VARCHAR(15) in the student table.

Q9. Drop the column dob from the student table.

Q10. Rename the column block to building in the department table.

Part B: Data Insertion (DML – INSERT)

Q11. Insert at least 3 records into the department table.

Q12. Insert at least 5 records into the student table.

Q13. Insert at least 4 records into the course table.

Q14. Insert a student record by specifying only selected columns.

Part C: Data Retrieval (DML – SELECT)

Q15. Display all records from the student table.

Q16. Display student name, email, and phone only.

Q17. Display students belonging to a specific department.

Q18. Display all courses having credit greater than 3.

Q19. Display students whose email is not NULL.

Part D: Data Modification (DML – UPDATE)

Q20. Update phone number of a student using student_id.

Q21. Change the department of a student.

Q22. Increase course credit by 1 for all courses of a specific department.

Part E: Data Deletion (DML – DELETE / TRUNCATE)

Q23. Delete a student record using student_id.

Q24. Delete all students belonging to a particular department.

Q25. Remove all records from the course table without deleting its structure.

Part F: Table & Database Removal (DDL)

Q26. Drop the course table.

Q27. Drop the student table.

Q28. Drop the department table.

Q29. Drop the engineering_college database.